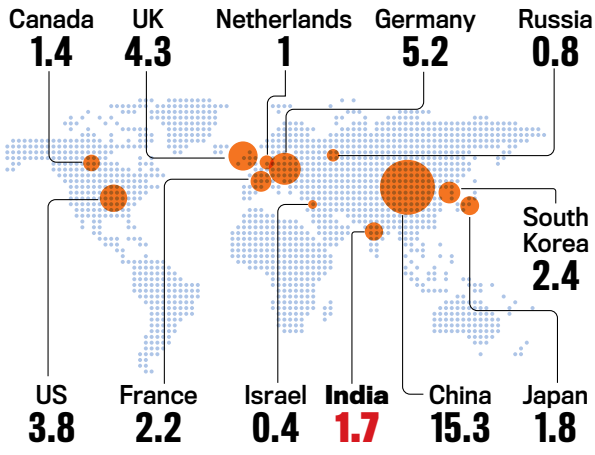
THE HI-TECH IMPERATIVES

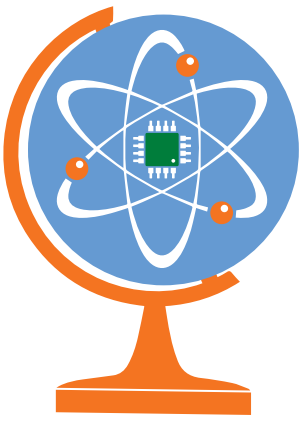
It all boils down to identifying and investing in promising opportunities in emerging technologies and creating a favourable ecosystem for private players to pitch in too

FUNDING BOOST

While China leads the pack in global public investments in quantum technology, India too is playing catch-up with at least the West

Note: The figures denote investments (in $ billion) announced by respective governments, as of 2023
Source: Quantum Technology Monitor, McKinsey & Co, April 2024

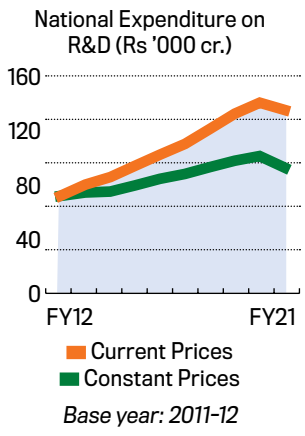




RENEWED FOCUS ON R&D

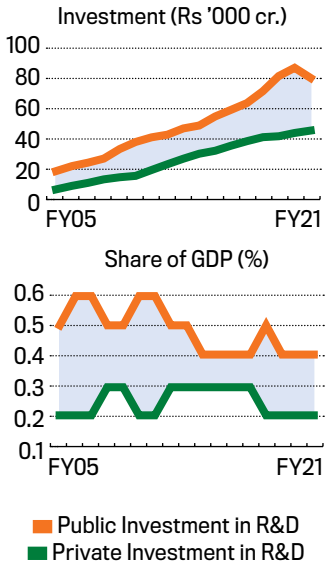
NATIONAL EXPENDITURE

India's spending on R&D saw a consistent rise. A dip in FY21 can be attributed to the Covid lockdowns



Base year: 2011-12

PUBLIC VS PRIVATE



WHERE INDIA STANDS IN GLOBAL R&D SPENDING

India's R&D expenditure may be just 0.6% of its GDP, but it's ahead of other major economies like the UK, Russia and Brazil in volume

R&D expenditure, 2020 (in bn current PPP$); as % of GDP in ( ) ● Per capita (current PPP$)

USA	721 (3.5%)	2,184
China	584 (2.4%)	413
Japan	174 (3.3%)	1,385
Germany	144 (3.1%)	1,736
Korea	113 (4.8%)	2,180
France	75 (2.3%)	1,097
India	**58 (0.6%)**	**42**
UK	56 (1.7%)	838
Russia	48 (1.1%)	327
Brazil	41 (1.3%)	198

Source: Department of Science and Technology; R&D Statistics, 2022-23

UNLIKE COMPUTER CHIP MANUFACTURING, THERE ARE HARDLY ANY COMMERCIAL FABRICATION FACILITIES FOR QUANTUM PROCESSORS GLOBALLY. IT'S AN OPPORTUNE TIME FOR INDIA TO START PLANNING FOR INVESTMENTS ON THAT FRONT